

GOD’S NUMBER OF THE $n \times n$ SLIDING PUZZLE IS $n^3 + O(n^2 \log n)$

ABSTRACT. Let $\mathcal{G}(n)$ be the largest distance, measured in single slides, between two mutually reachable configurations of the $n \times n$ sliding puzzle (the $(n^2 - 1)$ -puzzle). A simple potential argument shows $\mathcal{G}(n) \geq n^3 - O(n)$. We prove the matching upper bound $\mathcal{G}(n) \leq n^3 + O(n^2 \log n)$. The algorithm routes every tile from its starting cell up through a hierarchy of nested “star routers” to the centre of the board, and back down to its home. Tiles move in pipelined queues in which one slide advances one tile by one cell, so the total cost is twice the sum of the distances of all cells to the centre, which is at most n^3 , plus $O(n^2)$ for each of the $O(\log n)$ levels of the hierarchy.

1. INTRODUCTION

The $n \times n$ sliding puzzle has $n^2 - 1$ numbered tiles and one empty cell, the *blank*, on an $n \times n$ grid. A *move* (or *slide*) moves a tile adjacent to the blank into the blank. Two configurations are *connected* if one can be turned into the other by moves. *God’s number* $\mathcal{G}(n)$ is the maximum, over connected pairs, of the least number of moves joining them.

Theorem 1.1. *There are constants $C_\star$ and $N_\star$ such that for every $n \geq N_\star$, any two connected configurations of the $n \times n$ puzzle are joined by at most*

$$n^3 + C_\star n^2 (\log_{16} n + 1)$$

moves. Hence $\mathcal{G}(n) = n^3 + O(n^2 \log n)$.

The leading term cannot be improved.

Remark 1.2 (Lower bound). Fix a target board. A move changes the sum, over all non-blank tiles, of the Manhattan distance between a tile’s cell and its cell in the target by exactly one. Consider the board in which the tile whose target cell is x sits at the point reflection of x through the centre of the board. If it is not connected to the target, exchange two adjacent non-blank tiles: by the parity criterion of Johnson and Story [1] (two boards are connected exactly when the parity of the cell permutation between them equals the parity of the blank’s displacement) this makes it connected, and it changes the total by $O(1)$. Leaving out the blank’s own term, which is $O(n)$, the total is at least $\sum_{0 \leq i, i' < n} (|2i - n + 1| + |2i' - n + 1|) - O(n)$. The sum is n^3 for even n and $n^3 - n$ for odd n . Hence $\mathcal{G}(n) \geq n^3 - O(n)$.

The idea. Moving the blank once around a closed loop of cells shifts every tile on the loop by one cell, at a cost of one move per tile (Lemma 2.1). A long loop is therefore a conveyor belt that transports tiles at an amortised cost of one move per tile per cell. If every tile could travel from its start to the centre of the board and from the centre to its home along such belts, the total cost would be $2 \sum_x d(x, \text{centre}) \approx n^3$, where d is the Manhattan distance and x ranges over the cells.

To realise this we split the board recursively into a hierarchy. Each node of the hierarchy is a square. A non-leaf node is a *star router*: a small hub, and b^2 child squares (we take $b = 16$), each connected to the hub by a pair of conveyor belts (an export lane and a delivery lane). Tiles leave a child through its export lane and enter it through its delivery lane. A node talks to its parent through a narrow interface. It is fed one tile at a time and hands back one tile in exchange (Section 3). Each level of the hierarchy wastes only $O(T^2)$ moves at a node of side T , beyond the conveyor cost, and there are $O(\log n)$ levels. This gives $n^3 + O(n^2 \log n)$.

The difficulty is to make this precise. A node must hand out a tile before it has received the tiles that belong in it. It must keep its children supplied with exchange tiles. It must never get stuck. And it must account for every move. The interface of Section 3 isolates exactly what a node promises to its parent. The core of the paper (Section 5) shows that a star router whose children keep these promises keeps them too, with an explicit cost.

Plan of the paper. Section 2 collects elementary facts about moves. Section 3 defines *regions*, the interface every node implements. Section 4 treats the leaves of the hierarchy. Section 5 proves that a star router over regions is a region, and bounds its cost. Section 6 builds the hierarchy (the *corner tree*) and solves its cost recursion. Section 7 builds the root, a star over four quadrant trees around the centre of the board, and proves Theorem 1.1.

Formal verification. The complete argument, with explicit constants, has been formalised in the Lean 4 proof assistant with the Mathlib library. The formalisation uses only the standard axioms (propositional extensionality, choice and quotient soundness). This paper is self-contained and does not depend on it. We suppress all specific constants.

2. PRELIMINARIES

2.1. Boards and moves. Cells are pairs (i, i') with $0 \leq i, i' < n$ (row, column). Two cells are *adjacent* if they differ by one in exactly one coordinate. The distance $d(x, y)$ between $x = (i, i')$ and $y = (k, k')$ is the Manhattan distance $|i - k| + |i' - k'|$. A *board* is a bijection B from cells to tiles $\{0, 1, \dots, n^2 - 1\}$, where 0 is the blank; $\text{blank}(B) = B^{-1}(0)$. We say $B(x)$ is the *content* of cell x . A move exchanges the contents of $\text{blank}(B)$ and an adjacent cell. A *path* is a sequence of moves; its length is its number of moves. All costs in this paper are lengths of paths. We call a path *blank-returning* if the blank ends where it started.

A blank-returning path changes B into B' with $B'(\pi(x)) = B(x)$ for a permutation π of cells fixing $\text{blank}(B)$: the *effect* of the path. We say the path *moves the content of x to $\pi(x)$* . A 3-cycle $(a\ b\ c)$ moves the content of a to b , of b to c , and of c to a .

Lemma 2.1 (walks). *Let $c_0, c_1, \dots, c_k$ be distinct cells, each adjacent to the next, with the blank at c_0 . Moving the blank along them (cost k) moves the content of c_i to c_{i-1} for $1 \leq i \leq k$, and leaves the blank at c_k . If c_k is adjacent to c_0 and the blank continues to c_0 (total cost $k + 1$, a loop), the contents of $c_2, \dots, c_k$ move one step back along the loop and the content of c_1 moves to c_k . In both cases no other cell changes.*

Proof. Each move exchanges the blank with the next cell of the walk. □

Lemma 2.2 (conjugation). *Let σ be a path taking the blank from z to z' , and $\bar{\sigma}$ the same moves in reverse order. Let η be a blank-returning path from z' whose effect fixes every cell outside a set $U \ni z'$. Then $\sigma\eta\bar{\sigma}$ is blank-returning, has length $2|\sigma| + |\eta|$, and its effect fixes every cell x whose content σ does not carry into U . If moreover σ leaves every cell of $U \setminus \{z'\}$ with its content, then the effect of $\sigma\eta\bar{\sigma}$ agrees with that of η on $U \setminus \{z'\}$.*

Proof. Let $\hat{\sigma}$ be the permutation of cells, blank included, by which σ moves contents, and $\hat{\eta}$ that of η . Then $\bar{\sigma}$ acts by $\hat{\sigma}^{-1}$ and the composite acts by $\hat{\sigma}^{-1}\hat{\eta}\hat{\sigma}$, which fixes every x with $\hat{\sigma}(x) \notin U$. If $\hat{\sigma}$ fixes $U \setminus \{z'\}$ pointwise, then for $x \in U \setminus \{z'\}$ we have $\hat{\eta}(x) \in U \setminus \{z'\}$, because $\hat{\eta}$ fixes z' , so the composite maps x to $\hat{\eta}(x)$. □

The effect of a path on cells does not depend on the contents of the cells, only on the route of the blank. So it is harmless for a path to pass temporarily through cells used by some other structure, as long as its net effect restores them. We use Lemma 2.2 constantly in the following form: walk the blank along a walk, do an operation that changes only cells off the walk (other than the walk's end, where the blank is), and walk back. Every cell of the walk is then restored.

2.2. Local permutations. A *square* of side s is a set of cells $\{(r_0 + i, c_0 + j) : 0 \leq i, j < s\}$. A path stays *inside* a square if the blank never leaves it; such a path only changes cells of the square.

Lemma 2.3 (3-cycles). *Let $\mathcal{S}$ be a square of side $s \geq 6$ containing the blank, and let a, b, c be distinct non-blank cells of $\mathcal{S}$. There is a blank-returning path inside $\mathcal{S}$, of length $O(s)$, whose effect is the 3-cycle $(a\ b\ c)$. The same holds for $(a\ c\ b)$.*

Proof. Use coordinates $(i, i') \in [0, s)^2$ in $\mathcal{S}$. A *rectangle* is a set $[i_0, i_1] \times [j_0, j_1]$ of cells of $\mathcal{S}$.

Transport. Let U be a rectangle with both sides at least 3 that contains the blank and a tile t . Then t can be moved to any cell of U by $O(s)$ moves inside U . Indeed, to push t one step in some direction, bring the blank to the cell ahead of t without passing through t . This is possible because U minus one cell is connected, and it costs $O(s)$ the first time. Then exchange. To push again in the same direction, the blank, now just behind t , goes round t through a neighbouring row or column of U (one exists because the sides are at least 3) in 4 moves. At the bend of an L-shaped route the blank walks, inside U and avoiding t , to the cell ahead of t in the new direction ($O(s)$ moves). Moving t along an L-shaped route therefore costs $O(s)$. (This is the elementary step of the classical row-by-row algorithm; see e.g. [2].)

Staging. Let t_1, t_2, t_3 be the tiles at a, b, c . We place them at $(0, 0)$, $(0, 1)$, $(1, 1)$ in turn.

- (1) Transport t_1 to $(0, 0)$ inside $U = \mathcal{S}$.
- (2) Let $U' = [1, s-1] \times [0, s-1]$, which avoids $(0, 0)$. If t_2 is in row 0, bring the blank below it (avoiding $(0, 0)$ and t_2) and push t_2 down. Bring the blank into U' without moving t_2 or t_1 : if it is in row 0, walk it along row 0 to a column other than 0 and other than t_2 's (one exists since $s \geq 4$), and step down. Transport t_2 inside U' to $(1, 1)$. Move the blank, inside U' and then up through column 2, to $(0, 2)$, then to $(0, 1)$; this passes neither $(0, 0)$ nor $(1, 1)$. Push t_2 up into $(0, 1)$; the blank is now at $(1, 1)$.
- (3) In the same way, using U' , which avoids $(0, 0)$ and $(0, 1)$, transport t_3 to $(2, 1)$. If t_3 is in row 0, first push it down; when bringing the blank out of row 0, walk it only through columns ≥ 2 other than t_3 's. Bring the blank to $(1, 1)$ avoiding $(2, 1)$ and row 0, and push t_3 up. The blank is now at $(2, 1)$; move it to $(2, 0)$ and then to $(1, 0)$.

Each step costs $O(s)$ moves inside $\mathcal{S}$ and never disturbs the tiles already placed. Call the whole path σ . Afterwards $(0, 0)$, $(0, 1)$, $(1, 1)$ hold t_1, t_2, t_3 and the blank is at $(1, 0)$.

Gadget. Moving the blank round the block, $(1, 0) \rightarrow (0, 0) \rightarrow (0, 1) \rightarrow (1, 1) \rightarrow (1, 0)$, moves t_1 to $(1, 1)$, t_3 to $(0, 1)$ and t_2 to $(0, 0)$ (Lemma 2.1). Going round the other way does the inverse.

Conjugation. The path σ , then the gadget, then $\bar{\sigma}$, has the effect $\hat{\sigma}^{-1} \hat{\eta} \hat{\sigma}$ (Lemma 2.2), where $\hat{\eta}$ is the gadget's 3-cycle on the staged cells. This is a 3-cycle on a, b, c . The two directions of the gadget give the two 3-cycles $(a\ b\ c)$ and $(a\ c\ b)$. $\square$

Corollary 2.4 (double transpositions). *In the setting of Lemma 2.3, for distinct non-blank cells a, b, c, d of $\mathcal{S}$, the exchange of the contents of a and b together with that of c and d costs $O(s)$ moves, blank-returning, inside $\mathcal{S}$.*

Proof. Do the 3-cycle $(b\ c\ d)$, then $(a\ b\ c)$. $\square$

A single exchange of two contents is impossible with the blank returning (by the parity argument below). Every exchange we need is therefore paired with an exchange of two cells whose contents we do not care about. We call these two cells a *parity pair*.

Lemma 2.5 (local placement). *Let $\mathcal{S}$ be a square of side $s \geq 6$, and $\mathcal{R} \subseteq \mathcal{S}$ a set of cells containing the blank. Let $Z \subseteq \mathcal{R} \setminus \{\text{blank}\}$ with $|\mathcal{R}| \geq |Z| + 3$, and let ψ assign distinct non-blank tiles to the cells of Z , each currently in a cell of $\mathcal{R}$. Then a blank-returning path of length $O(s|Z|)$ makes every $f \in Z$ contain $\psi(f)$, and changes no cell outside $\mathcal{R}$.*

Proof. Fix two cells $y_1 \neq y_2$ of $\mathcal{R} \setminus (Z \cup \{\text{blank}\})$. Treat the cells of Z one at a time. If f does not yet hold $\psi(f)$, let y be the cell holding $\psi(f)$. Then y is not the blank cell, and it is not a cell treated earlier, since ψ is injective. Pick $y' \in \{y_1, y_2\}$ with $y' \neq y$ and perform the 3-cycle $(y \ f \ y')$ by Lemma 2.3. This places $\psi(f)$ in f and changes only y, f, y' , none of them treated earlier. $\square$

2.3. Parity and global clean-up.

Lemma 2.6 (parity). *If B and B' are connected and have their blank in the same cell, then the permutation of cells carrying contents of B to contents of B' is even.*

This is the classical criterion of Johnson and Story [1]; we only need the easy direction.

Proof. Colour the cells like a chessboard. Every move changes the colour of the blank cell, so a path from B to B' has even length. Each move is a transposition of two contents, so the permutation is a product of an even number of transpositions. $\square$

Lemma 2.7 (clean-up). *Let $n \geq 6$, and let B, B' be connected boards with the same blank cell that differ on a set S of cells. Then they are joined by a path of length $O(n|S|)$.*

Proof. The permutation φ carrying B 's contents to B' 's is supported on S and is even (Lemma 2.6). If φ has a cycle of length at least 3, one 3-cycle places two of its contents correctly. Otherwise φ is a product of at least two disjoint transpositions, and two 3-cycles fix four cells (Corollary 2.4). By Lemma 2.3, applied with the whole board as the square, each 3-cycle costs $O(n)$; it avoids the blank cell, which is already correct. The support shrinks by at least one per 3-cycle. $\square$

Lemma 2.8 (gathering). *Let $n \geq 6$. Let $\mathcal{K}$ be a set of k tiles containing 0, and $\mathcal{R}$ a set of k cells containing the blank of B , with $2k < n^2$. A path of length $O(nk)$, keeping the blank cell, makes the tiles of $\mathcal{K}$ occupy exactly the cells of $\mathcal{R}$.*

Proof. While some tile of $\mathcal{K}$ lies outside $\mathcal{R}$, say in x , some cell $y \in \mathcal{R}$ holds a tile not in $\mathcal{K}$. Pick a cell $y' \notin \mathcal{R} \cup \{x\}$; it is not the blank cell, and it exists since $2k < n^2$. The 3-cycle $(x \ y \ y')$ costs $O(n)$ and moves one more tile of $\mathcal{K}$ into $\mathcal{R}$. $\square$

Lemma 2.9 (blank walk). *The blank can be moved to any cell c in at most $2n$ moves, changing only cells in the blank's current column and in the row of c .*

Proof. Walk vertically, then horizontally (Lemma 2.1). $\square$

3. REGIONS: THE INTERFACE OF A NODE

Fix once and for all a *goal board* G . Every node of the hierarchy works towards G : the tile $G(x)$ belongs in cell x .

3.1. Frames.

Definition 3.1 (frame). A *frame* F consists of:

- three pairwise disjoint sets of cells: the *core homes* $\text{Core}(F)$, the *reserve homes* $\text{Res}(F)$ and the *spare cells* $\text{Spare}(F)$, with $\text{cells}(F)$ their union and $G(x) \neq 0$ for $x \in \text{Core}(F) \cup \text{Res}(F)$;
- two further cells $e_F \neq z_F$ outside $\text{cells}(F)$, the *entry pair*: the *input cell* e_F and the *blank cell* z_F .

We write $M = |\text{Core}(F)|$, $r = |\text{Res}(F)|$, $g = |\text{Spare}(F)|$. The *core goals* are the tiles $G(\text{Core}(F))$ and the *reserve goals* are $G(\text{Res}(F))$. A tile is a *goal* of F if it is a core or reserve goal. All our frames have $g \geq 6$.

The reason for two kinds of home will appear in Section 5. In a star router, core homes are homes inside children, which are reached through the conveyor belts. Reserve homes are the router's own cells, which are filled by direct exchanges.

3.2. Roles and requests. For a non-blank tile t we call $G^{-1}(t)$ the *home* of t . A region (defined precisely in Definition 3.3) is operated by another party, its *caller*, through requests; in our construction the caller is the region's parent in the hierarchy, and the parent's caller, and so on, are its *ancestors*. Between requests the region is *paused*: the blank is not in its cells. The region keeps a *record* assigning each tile currently in $\text{cells}(F)$ exactly one *role*:

core source, reserve source, helper, core final, reserve final.

Sources are tiles that were in the region at the start and have not yet been handed out. Finals are goal tiles handed to the region with the information that they are goals. Helpers are everything else. Four *counters* record history: e_c, e_r count the core and reserve sources handed out, and a_c, a_r count the core and reserve finals received. At the *start*, every tile in a core home is a core source, every tile in a reserve home a reserve source, and every tile in a spare cell a helper; all counters are 0.

A *request* is made with the blank at z_F and an *input* tile t at e_F , with an *arrival kind* $\alpha \in \{\text{core}, \text{reserve}, \text{helper}\}$ such that t is a core goal, a reserve goal, or any non-blank tile, respectively. Core and reserve arrivals are *goal arrivals*. The region answers with a path, after which the tile at e_F , the *output*, is a tile that had the role $\text{sel}(\alpha)$ given by the table below; it leaves the record, t joins the record as a core final, reserve final or helper (for $\alpha = \text{core}, \text{reserve}, \text{helper}$), and the counters are updated.

The selected role follows the *source-first policy*: hand out a source while any remain, preferring reserve sources for reserve arrivals.

arrival α	output role $\text{sel}(\alpha)$
helper	core source if $e_c < M$, else reserve source
core	core source if $e_c < M$, else reserve source if $e_r < r$, else helper
reserve	reserve source if $e_r < r$, else core source if $e_c < M$, else helper

A helper request is only *allowed* while some source remains ($e_c < M$ or $e_r < r$). A goal arrival is always allowed. The input sits outside the region, so it is not one of the region's finals (finals never leave a region; see the remark after Definition 3.3); since the finals are distinct goals, $a_c < M$ for a core arrival and $a_r < r$ for a reserve arrival.

Lemma 3.2 (counting). *Under the source-first policy the counters always satisfy $e_c \leq M$, $e_r \leq r$, $a_c \leq M$, $a_r \leq r$, $a_r \leq e_r$ and $a_c + a_r \leq e_c + e_r$. The record holds $M - e_c$ core sources, $r - e_r$ reserve sources, a_c core finals, a_r reserve finals and $g + \lambda$ helpers, where*

$$\lambda = (e_c + e_r) - (a_c + a_r) \geq 0$$

is the lead. In particular the selected role is never empty, and when $a_c = M$ and $a_r = r$ every source has been handed out.

Proof. The core finals are distinct core goals, so $a_c \leq M$; likewise $a_r \leq r$. A reserve arrival hands out a reserve source while any remain; once $e_r = r$, $a_r \leq r = e_r$. A goal arrival hands out a source unless none remain, in which case $e_c + e_r = M + r \geq a_c + a_r$. The role counts follow from $|\text{cells}(F)| = M + r + g$. In particular there are always at least $g \geq 6$ helpers, and a selected source role is non-empty by the condition in the table. $\square$

The lead is the number of tiles the region has handed out in excess of the goals it has received. A request changes it by +1 (helper arrival), 0 (goal arrival answered with a source) or -1 (goal arrival answered with a helper). The region must hold that many extra helpers, and its routing

cost grows with the lead, so the caller promises a bound: an *allowance* β with $\lambda \leq \beta$ after every request.

3.3. Regions. A region is a state machine. Its state carries the record, the counters, and a number ω , the *work* done so far (the total length of all paths it has produced), plus whatever private data it needs.

Definition 3.3 (region). A *region* consists of a frame F (Definition 3.1), two numbers base and rate (its *budget*), a set of *states* (each with a record, counters and work ω), and a relation “state ς is *valid on board* B *with allowance* β ” satisfying the following laws.

- (R1) *Soundness.* If ς is valid on B , the blank is not in $\text{cells}(F)$, and the tiles in $\text{cells}(F)$ are exactly the tiles of the record.
- (R2) *Start.* For every board B whose blank is not in $\text{cells}(F)$ there is a state valid on B with allowance 0, with the start record, zero counters and $\omega = 0$.
- (R3) *Raising the allowance.* Valid with allowance β implies valid with every $\beta' \geq \beta$.
- (R4) *Pausing.* Validity depends only on the contents of $\text{cells}(F)$: if B, B' agree on $\text{cells}(F)$, then ς is valid on B iff it is valid on B' (same allowance). So the caller may change any other cell between requests.
- (R5) *Substitution.* Let ς be valid on B with allowance β , and let π be a permutation of tiles fixing 0 and every tile whose role in ς is not helper. If $B'(x) = \pi(B(x))$ for $x \in \text{cells}(F)$, then some state ς' is valid on B' with allowance β , with the record relabelled by π , the same counters and the same work. Physically, an outsider replaces some helpers by other tiles.
- (R6) *Requests.* Let ς be valid on B with allowance β , with the blank at z_F , and let an allowed request be made whose resulting lead is at most β . Then there are a path from B to some B' and a state ς' valid on B' with allowance β , such that the blank is back at z_F , every cell outside $\text{cells}(F) \cup \{e_F\}$ has its previous content (cells may be disturbed in between), the output, record and counters change as described above, and $\omega(\varsigma') \geq \omega(\varsigma) +$ the path's length.
- (R7) *Finish.* If ς is valid on B with allowance β , the blank is at z_F , $a_c = M$ and $a_r = r$, then there is a blank-returning path after which every $x \in \text{Core}(F) \cup \text{Res}(F)$ contains $G(x)$, no cell outside $\text{cells}(F)$ has changed, and $\omega(\varsigma)$ plus its length is at most $\text{base} + \text{rate} \cdot \beta$.

Finals never leave a region: sel never selects a final, substitutions fix non-helpers, and a request changes only $\text{cells}(F) \cup \{e_F\}$. This justifies the remark that a goal input is not one of the region's finals. So the total work of a region over its whole life, finishing included, is at most $\text{base} + \text{rate} \cdot \beta$, where β is the largest allowance granted. Helpers may be arbitrary tiles, even goal tiles of the region that have not yet been declared as goals. The substitution law lets an ancestor take a helper out of a region and put another tile in its place, without the region noticing anything that matters to it.

4. LEAVES

Proposition 4.1. *Let F be a frame whose cells, entry pair included, lie in a square $\mathcal{S}$ of side $m \geq 6$, with $g \geq 6$. Then F is a region with $\text{base} = O(m^3)$ and $\text{rate} = 0$.*

Proof. A state is a record, counters and work ω . It is valid on B if (R1) holds, the counters are those of the history, and $\omega \leq cm(e_c + e_r + a_c + a_r)$ for a suitable constant c . Positions of tiles inside $\text{cells}(F)$ play no role, so (R2)–(R5) are immediate. A request selects any tile of role $\text{sel}(\alpha)$, sitting in some cell q , and exchanges the contents of e_F and q together with a parity pair of two other cells of $\text{cells}(F)$ (Corollary 2.4 in $\mathcal{S}$, which contains the blank at z_F). This costs $O(m)$ and keeps every tile of the record inside $\text{cells}(F)$. There are at most $e_c + e_r + a_c + a_r \leq 2(M + r) \leq 2m^2$

requests. Finishing is Lemma 2.5 with $\mathcal{R} = \text{cells}(F) \cup \{z_F\}$ and $Z = \text{Core}(F) \cup \text{Res}(F)$: by then all goals are in the region, and the $g \geq 6$ spare cells provide the room. It costs $O(m \cdot m^2)$. $\square$

5. STAR ROUTERS

This section proves the main structural result: a star router whose children are regions is itself a region, with an explicit budget (Theorem 5.11).

5.1. Layout.

Definition 5.1 (star layout). A *star layout* P with J *spokes* consists of the following, all inside a square $\mathcal{S}_P$ of side $s_P \geq 6$ (the *square* of P):

- J regions $Q_1, \dots, Q_J$, the *children*, with pairwise disjoint cells;
- distinguished *hub cells*: the *carrier cell* K , three *parity cells* u, v, u' , the *park* w , and one cell H_j for each spoke. They lie in a square of side $h \geq 6$, the *hub square*. The hub square may contain other cells, even cells of children. We only ever use it for double transpositions of hub cells, which change nothing else;
- for each spoke j , a *loop*: a cyclic sequence of distinct cells

$$L_j : \quad x_j, \underbrace{\varepsilon_{j,1}, \dots, \varepsilon_{j,p_j}}_{X_j}, \underbrace{e_j, \delta_{j,1}, \dots, \delta_{j,q_j}}_{Y_j}, H_j$$

each cell adjacent to the next and H_j adjacent to x_j , with $\varepsilon_{j,p_j} = z_j$ and (e_j, z_j) the entry pair of Q_j . The *export queue* X_j runs from the hub end (its *head* $\varepsilon_{j,1}$) to the child end z_j . The *delivery queue* Y_j runs from the child end (its head e_j) towards the hub. We also call X_j and Y_j the export and delivery *lanes*;

- for each spoke an *access walk* A_j from w to x_j ;
- six *marker* cells, and the *entry pair* (e_P, z_P) of P with a *connector* walk from z_P to w ;
- a square of side $h_w \geq 6$, the *wrapper square*, containing e_P, z_P, w, K, u, v .

These satisfy the following incidence conditions:

- loops are pairwise disjoint and avoid all children, w, K, u, v, u', e_P, z_P and the markers;
- A_j meets L_j only in x_j and avoids K, u, v and Q_j ;
- the connector avoids loops, children, K, u, v and e_P ;
- K, u, v, u', w and the markers lie in no child.

Finally, a set $\text{Body}(P) \subseteq \mathcal{S}_P$, not containing e_P, z_P , contains the children, the hub cells, the loops, the access walks and the markers. The cells of the body outside the children and the markers are the *own cells* $\text{Own}(P)$ of P . They include all loops and hub cells. The connector may leave the body.

The frame of P (Definition 3.1) is

$$\text{Core}(P) = \bigcup_j (\text{Core}(Q_j) \cup \text{Res}(Q_j)), \quad \text{Res}(P) = \text{Own}(P) \cup \bigcup_j \text{Spare}(Q_j), \quad \text{Spare}(P) = \text{markers},$$

with entry pair (e_P, z_P) . So $\text{cells}(P) = \text{Body}(P)$. The children's spare cells, which hold their helpers, are reserve homes of P . Write $\text{pop}_j = |\text{Core}(Q_j)| + |\text{Res}(Q_j)|$, the number of homes in child j .

5.2. The view from P . Each tile in $\text{cells}(P)$ has a role in P 's record. The core sources and core finals of P are its *core tiles*; every other tile of P (helpers, reserve sources, reserve finals) is a *token*. P 's routing only distinguishes core sources, core finals and tokens.

The core sources of P are the tiles initially in children's homes, which are exactly the sources of the children. They leave a child only as the child's outputs, travel along the child's export queue to the hub, and are handed out by P . A core final of P is a goal tile whose home is in

some child j , its *home spoke*. It travels along Y_j and enters Q_j as a core or reserve arrival of Q_j , according to its home. Tokens enter children only as helper arrivals, or through a direct exchange (a substitution for the child, Section 5.5).

5.3. Services. The basic operation is a *service* on spoke j . Services are performed with the blank at the park w , and the tile in K at that time is called the *carrier*. The *last place* of X_j is z_j , and the last place of Y_j is δ_{j,q_j} , the cell next to H_j .

The child call. In step 3 below, let t' be the tile at e_j . If t' is a core final of P , it is a goal of Q_j that is not in Q_j , and Q_j is called with arrival kind core or reserve according to its home. If t' is a token, it is passed to Q_j as a helper arrival if Q_j still has a source (so the request is allowed). Otherwise no request is made (an *idle turnaround*).

Lemma 5.2 (service). *Suppose the blank is at w and child j is paused. A service on spoke j is the following path:*

- (1) *exchange the contents of K and H_j , together with the parity pair (u, v) , inside the hub square (Corollary 2.4);*
- (2) *walk the blank along A_j to x_j , then around the loop to z_j ;*
- (3) *make the child call;*
- (4) *continue around the loop, $z_j \rightarrow e_j \rightarrow \delta_{j,1} \rightarrow \dots \rightarrow \delta_{j,q_j} \rightarrow H_j \rightarrow x_j$, and walk back along A_j to w ;*
- (5) *exchange K and H_j , together with (u, v) , again.*

Apart from changes inside Q_j made by the child's request, its net effect is:

- *the new carrier is the old head of X_j ;*
- *the rest of X_j moves one place towards the head, and the last place of X_j receives the child's output (after an idle turnaround, the tile t' itself);*
- *the tile t' at the head of Y_j is the child's input, the rest of Y_j moves one place towards the head, and the last place of Y_j receives the old carrier.*

Every other cell, including H_j , u , v , w , the access walk, the connector and the other loops, ends with its previous content. The cost is $\text{cost}_j = |L_j| + 2|A_j| + O(h)$ plus the cost of the child's request, where $|L_j|$ is the number of cells of the loop and $|A_j|$, like the length of the connector, is the number of moves along the walk.

Proof. Step 1 puts the carrier into H_j . In step 2, by Lemma 2.1, the walk along X_j moves each content of X_j one place towards x_j , the content of x_j having been pushed onto the access walk. When the blank is at z_j the child may be called: its request needs exactly the blank at z_j and the input at e_j , and changes only cells $(Q_j) \cup \{e_j\}$ (net). Step 4 moves the output from e_j to z_j , shifts Y_j one place towards e_j , moves the carrier from H_j into the last place of Y_j , and moves the old head of X_j (now at x_j) into H_j . Walking back along A_j restores the access walk and x_j by Lemma 2.2, applied with $U = L_j \cup \text{cells}(Q_j)$: the access walk meets U only in x_j (it meets L_j only there and avoids Q_j). By the lemma's second clause, the loop and the child change exactly as the loop walk and the child call alone would change them. Step 5 moves the old export head to K , returns H_j 's original content, and swaps u, v back. $\square$

The two queues of a spoke thus behave as first-in first-out queues joined at the hub and at the child. One service advances every tile in them by one place, at a cost of $|L_j| + O(h + |A_j|)$: about one move per tile per cell.

5.4. The state of P .

Definition 5.3 (state of a star). A state of P consists of its record, counters e_c, e_r, a_c, a_r and work ω ; a state ς_j of each child; and the following numbers, which form the *ledger*. For each spoke j :

- out_j , the number of core sources of P brought to the carrier from X_j ; in_j , the number of core finals of P sent into Y_j as carriers; the *credit* $C_j = \text{out}_j - \text{in}_j$;
 - the *peak* $\hat{C}_j$, the largest value $\max(0, C_j)$ has had so far;
 - k_j , the number of services on spoke j , split as $k_j = \text{out}_j + \text{tok}_j + \text{late}_j + \text{dr}_j$ according to what the service brought to the carrier and when: a core source; a token while j was active; anything, with a core final carrier, while j was retired; anything, with a token carrier, while j was retired;
 - for a retired spoke, the number t_j of active spokes just before the service that retired it.
- Spoke j is *active* while $\text{out}_j < \text{pop}_j$ and *retired* afterwards (a spoke with $\text{pop}_j = 0$ is retired from the start). Finally, the state records the numbers n_{req} of wrappers and n_{dx} of direct exchanges performed (these operations are defined in Section 5.5), and the number n_{late} of core arrivals received while $e_c = M$.

P grants child j the allowance

$$(1) \quad \beta_j = \hat{C}_j + |X_j| + |Y_j| + 2,$$

which never decreases. Number the places of X_j as 1 (the head) to $p_j = |X_j|$ (the last place). A tile in place i has travelled from the last place if $k_j \geq p_j + 1 - i$; we then call place i *refilled*. Write $\mathbf{a}$ for the number of active spokes and J' for the number of spokes with $\text{pop}_j \geq 1$.

Definition 5.4 (validity). A state of P is *valid* on a board B with allowance β if the following hold, where “the carrier” means the tile in K . All items are required. The items marked $\dagger$ concern the *rest* situation, between requests, when the blank is outside $\text{cells}(P)$; in the middle of a request or of the finish only the remaining items, the routing conditions defined below, are maintained.

- (I0) $\dagger$ The tiles in $\text{cells}(P)$ are exactly the tiles of the record, and the blank is not in $\text{cells}(P)$.
- (I1) Each child state ς_j is valid on B with allowance β_j .
- (I2) Every core source of P is a source (core or reserve) of some child, or lies in some export queue, or, only during a request, is the carrier. Conversely every source of a child is a core source of P .
- (I3) Every core final of P is a final of some child, or lies in the delivery queue of its home spoke, or, only during a request, is the carrier. Conversely every final of a child is a core final of P . (So at rest the carrier is a token.)
- (I4) Delivery queues contain no core source of P ; export queues contain no core final of P .
- (I5) For each j , writing $e^{(j)}, a^{(j)}$ for the totals of handed-out sources and received finals of Q_j :

$$e^{(j)} = \text{out}_j + \#\{\text{core sources of } P \text{ in } X_j\}, \quad a^{(j)} + \#\{\text{core finals of } P \text{ in } Y_j\} = \text{in}_j.$$

- (I6) $\sum_j \text{out}_j$ and $\sum_j \text{in}_j$ are the numbers of core sources brought to the carrier and of core finals sent from it; $\dagger$ they equal e_c and a_c .
- (I7) If a refilled place of X_j holds a token, then Q_j has no source and every place $i' > i$ of X_j (nearer the child) holds a token.
- (L1) Retired spokes have $C_j \geq 0$. Every active spoke has $\hat{C}_j \leq 1 + \beta/\mathbf{a}$. Every retired spoke with $\text{pop}_j \geq 1$ has $\hat{C}_j \leq 1 + \beta/t_j$, and these t_j are distinct numbers in $\{\mathbf{a} + 1, \dots, J'\}$. Spokes with $\text{pop}_j = 0$ have $\hat{C}_j = 0$.
- (L2) $\text{tok}_j \leq \min(k_j, |X_j|)$; $\text{late}_j = 0$ while j is active, and $\text{late}_j + \text{pop}_j \leq \text{in}_j + \hat{C}_j$ once it is retired; $\dagger$ $\text{dr}_j = 0$.
- (L3) $\dagger$ P 's counters satisfy Lemma 3.2 with lead $\lambda \leq \beta$; $n_{\text{req}} \leq e_c + e_r + a_c + a_r$; $n_{\text{dx}} \leq a_r + e_r + n_{\text{late}}$; $n_{\text{late}} = 0$ if $e_c < M$, and $n_{\text{late}} + M - a_c \leq \beta$ if $e_c = M$.

(L4) $\omega \leq \sum_j k_j \text{cost}_j + \sum_j \omega(\zeta_j) + n_{\text{req}} c_{\text{wrap}} + n_{\text{dx}} c_{\text{dx}}$, with c_{wrap} and c_{dx} as in Section 5.5. The items without $\dagger$, together with the non- $\dagger$ part of (L2), are the *routing conditions*. They are meaningful in the middle of a request or of the finish, when the blank is at w .

At the start every count is 0, the queues and K hold own-cell tiles (reserve sources of P , so tokens), no place is refilled, every spoke with $\text{pop}_j \geq 1$ is active with $\hat{C}_j = 0 \leq 1$, and each child starts by (R2). So the start state is valid with allowance 0.

Lemma 5.5 (consequences). *In a valid state:*

- (1) *the lead of child j is at most $C_j + |X_j| + |Y_j| \leq \beta_j - 2$;*
- (2) $\sum_j \hat{C}_j \leq J + \beta \mathcal{H}_J$, *where $\mathcal{H}_J = 1 + \frac{1}{2} + \dots + \frac{1}{J}$;*
- (3) $k_j \leq \text{pop}_j + |X_j| + \hat{C}_j + \text{dr}_j$.

Proof. (1) Subtract the two equations of (I5). (2) By (L1), the active spokes contribute at most $a(1 + \beta/a)$, and the retired spokes with $\text{pop}_j \geq 1$ at most $\sum_{t=a+1}^{J'} (1 + \beta/t)$. If $a \geq 1$ the total is $J' + \beta(\mathcal{H}_{J'} - \mathcal{H}_a + 1) \leq J + \beta \mathcal{H}_J$, as $\mathcal{H}_a \geq 1$. If $a = 0$ it is $J' + \beta \mathcal{H}_{J'}$. (3) $\text{out}_j \leq \text{pop}_j$ and $\text{tok}_j \leq |X_j|$, and $\text{late}_j \leq \hat{C}_j$ because $\text{in}_j \leq \text{pop}_j$ (the core finals sent along Y_j are distinct goals of Q_j). $\square$

5.5. Requests. Let P be at rest with the blank at z_P . A request brings an input t at e_P with kind α . P uses two kinds of operation, the wrapper and the direct exchange, which we describe now; Table 1 below says which is used when.

The wrapper. Walk the blank along the connector to w . Exchange the contents of e_P and K , together with the parity pair (u, v) , inside the wrapper square. Now the input is the carrier, and e_P holds the old content of K . Run some services (the *inner run*, below). Exchange e_P and K with (u, v) again, putting the final carrier at e_P as output. Walk back. Every service restores e_P , the connector, u and v (Lemma 5.2), so the wrapper changes nothing outside $\text{cells}(P) \cup \{e_P\}$. The wrapper costs $c_{\text{wrap}} = O(h_w + |\text{connector}|)$ plus its services.

The direct exchange. With the blank at z_P , exchange the content of e_P with that of some cell $q \in \text{cells}(P)$, together with a parity pair chosen among u, v, u' avoiding q . This is a double transposition inside $\mathcal{S}_P$, which contains the blank, and costs $c_{\text{dx}} = O(s_P)$ (Corollary 2.4). We only exchange a token at e_P with a token at q . The parity cells are own cells off the loops and K , and they hold tokens. So no core tile of P moves, and (I2)–(I7) and the ledger are unaffected ((I7) only cares whether a place holds a token). If q lies in a child Q_j , its token is a helper of Q_j , because by (I2) and (I3) the non-helpers of Q_j are core tiles of P . The exchange is then a substitution for Q_j , so Q_j stays valid by (R5).

Scheduling rule. During a request, a carrier that is a core final of P is served on its home spoke, and a carrier that is a token is served on an active spoke of least credit. The only other services are those of the *drain* at the end (Lemma 5.10), which serve token carriers on retired spokes.

The inner run. With a carrier in K :

- if the carrier is a core final of P , perform one service on its home spoke;
- then, while the carrier is a token and some spoke is active, perform a *helper round*: one service on an active spoke of least credit.

By (L2), which services keep (Lemma 5.7 below), the active spokes bring at most $\sum_j |X_j|$ tokens in all, and every other service on an active spoke brings a core source. So the inner run ends.

Bookkeeping during a request. When an operation brings the input into $\text{cells}(P)$ (into K by the wrapper, or into a cell q by a direct exchange), the input joins P 's record: as a core final for a core arrival, as a reserve final for a reserve arrival, and as a helper otherwise. The tile that the wrapper moves from K to e_P stays in the record while it waits there. P 's counters e_c, e_r, a_c, a_r

are updated, and the output leaves the record, only at the end of the request. The ledger is updated at the end of each service (its step 5). The child call in step 3 uses the allowance β_j as it was before the service; β_j never decreases, so the child stays valid afterwards.

Lemma 5.6 (credit cap). *Before every service, $\sum_j C_j \leq \beta$.*

Proof. $\sum_j C_j = \sum_j \text{out}_j - \sum_j \text{in}_j$. Within a request, services run only while the carrier is not a core source, so before each service $\sum_j \text{out}_j = e_c$ and $\sum_j \text{in}_j \geq a_c$, with the counters as before the request. By Lemma 3.2, $e_c - a_c = \lambda + (a_r - e_r) \leq \lambda \leq \beta$. During the drain, $a_c = M = e_c$ and drain services change no out_j or in_j , so $\sum_j C_j = 0$. $\square$

5.6. Services keep the routing conditions.

Lemma 5.7 (service preserves validity). *Suppose the routing conditions hold with allowance β , the blank is at w , and a service is made on a spoke chosen by the scheduling rule at a moment when $\sum_j C_j \leq \beta$. Then afterwards the routing conditions hold for the updated state, with allowance β .*

Proof. Lemma 5.2 describes the moves. We follow the four tiles that change place.

- *The child's input t'* , from the head of Y_j . If it is a core final of P , its home is in Q_j by (I3), the request is an allowed goal arrival, and t' becomes a final of Q_j . If it is a token passed as a helper arrival, Q_j has a source, so the request is allowed. The child's lead after the request is at most its lead before plus one, so it is at most β_j by Lemma 5.5(1). By (R6) for Q_j the child stays valid. Nothing outside $Q_j \cup \{e_j\}$ changes net.
- *The child's output* (or t' itself, after an idle turnaround) enters the last place of X_j . It is a source of Q_j , and so a core source of P , if Q_j had a source when it was called. Otherwise it is a token, and in that case Q_j has no source now or later. In particular a core final never enters X_j .
- *The old head of X_j* becomes the carrier. If it is a core source, out_j increases.
- *The old carrier* enters the last place of Y_j . If it is a core final, j is its home spoke by the scheduling rule, and in_j increases. A core source is never a carrier at a service: within a request, services stop as soon as the carrier is a core source (see the inner run in Section 5.5).

This keeps (I2)–(I6). For (I5) in detail: a source output adds one to $e^{(j)}$ and one core source to X_j ; a core source brought from the head adds one to out_j and removes one from X_j ; a core final carrier adds one to in_j and one core final to Y_j ; a core final entering Q_j moves one from Y_j to $a^{(j)}$; tokens change none of these. For (I7), places move one step towards the head and the new last place is refilled. A token arriving at the last place comes from a source-less child, and then no place is behind it. If an earlier refilled place holds a token, the child was already source-less, so the new last tile is also a token.

For the ledger, a core final carrier raises in_j by one and out_j by at most one, so no credit increases. A token carrier is served on an active spoke of least credit, or during the drain, where nothing in out, in changes. Retired spokes have $\text{out}_j = \text{pop}_j$ fixed and $\text{in}_j \leq \text{pop}_j$, so their credits never increase and stay nonnegative. Just before a token service on an active spoke, the active credits sum to at most β , since the total is at most β and the retired credits are nonnegative. Their minimum is at most their average, so the chosen credit is at most β/a , and afterwards at most $1 + \beta/a$. Since a never increases, the bound $1 + \beta/a$ is never smaller than earlier bounds, so (L1) holds for active spokes. If the service retires spoke j , we set $t_j = a$ (the old count). Its peak obeys $\hat{C}_j \leq 1 + \beta/t_j$ and never changes again. The new a is $t_j - 1$, which keeps the t 's distinct and above a .

For (L2), suppose the service is on an active spoke and brings a token, and suppose $k_j \geq p_j$ before it, so the head place is refilled. By (I7), Q_j has no source and X_j holds no core source, so by (I5) $\text{out}_j = e^{(j)} = \text{pop}_j$ and the spoke was already retired, a contradiction. Hence such a service has $k_j < p_j$, which keeps $\text{tok}_j \leq \min(k_j, |X_j|)$. When a spoke retires, $\text{late}_j = 0$ and $\text{pop}_j - \text{in}_j = C_j \leq \hat{C}_j$; each later late service raises both late_j and in_j by one. For (L4): the service's own moves cost cost_j , k_j increases by one, and the child's work increases by at least the length of its request by (R6). $\square$

case	condition	action
Ca	core, $e_c < M$	wrapper: service on home spoke, then helper rounds; output a core source
Cb	core, $e_c = M$	wrapper: one service on home spoke; then possibly one direct exchange
Ha	helper, $e_c < M$	wrapper: helper rounds; output a core source
Hb	helper, $e_c = M$	direct exchange with a reserve source
Ra	reserve, $e_r < r$	direct exchange with a reserve source
Rb	reserve, $e_r = r$, $e_c < M$	direct exchange with a helper, then wrapper as in Ha
Rc	reserve, $e_r = r$, $e_c = M$	direct exchange with a helper

TABLE 1. How a star router answers a request.

5.7. Answering requests.

Lemma 5.8 (requests). *Let the state be valid on B with allowance β , with the blank at z_P , and let an allowed request be made whose resulting lead is at most β . Answering it as in Table 1 gives a path and a new state, valid with allowance β , that satisfy law (R6).*

Proof. The output has role $\text{sel}(\alpha)$.

- Ca, Ha: the inner run ends only when the carrier is a core source, or when no spoke is active. But $e_c < M$ means some core source of P has not reached the carrier; by (I2) it is in a child or an export queue, so its spoke is active. So the output is a core source.
- Cb: since $e_c = M$, every spoke is retired, and by (I2) and (I4) the export head of the home spoke is a token. So the carrier y after the single service is a token. If y has role $\text{sel}(\alpha)$ (a reserve source if $e_r < r$, else a helper), it is the output. Otherwise exchange it directly with a tile of the right role. Every such tile is in $\text{cells}(P)$, since y is at e_P , and there is one by Lemma 3.2.
- Hb, Ra, Rc: one direct exchange with a tile of the selected role. The input is parked in the vacated cell as a helper or reserve final.
- Rb: exchange the reserve goal t directly with a helper, parking t as a reserve final, then treat the helper at e_P as in Ha. The output is a core source, which is $\text{sel}(\text{reserve})$ when $e_r = r$ and $e_c < M$.

Validity. At the start of the request the state is valid, so the routing conditions hold. Services keep them by Lemmas 5.7 and 5.6, and direct exchanges keep them, as shown above (the routing conditions only say where tiles are, so they make sense with the blank at z_P as well). For (I0) at the end: every step of the request (walks, services, double transpositions, the child calls) has a net effect that only permutes the contents of $\text{cells}(P) \cup \{e_P\}$. In net, the input has entered $\text{cells}(P)$ and the output has come out to e_P , which is exactly how the record changes. The blank is back at z_P . No drain occurs, so dr_j stays 0. The wrapper moves the input into K and the

final carrier out to e_P . The input is a core final (Ca, Cb), and then it is counted in a_c , or it is a token. The output is a core source counted in e_c , or a token. So the items marked $\dagger$ hold again at the end. For (L3), each request adds at most one wrapper and adds at least one to $e_c + e_r + a_c + a_r$. A direct exchange occurs with a reserve arrival (Ra–Rc: a_r grows), with a reserve source handed out (Hb, and Cb when $e_r < r$: e_r grows), or with a core arrival while $e_c = M$ (Cb when $e_r = r$: n_{late} grows), so $n_{\text{dx}} \leq a_r + e_r + n_{\text{late}}$ is kept. When e_c first reaches M , the new lead is $M + e_r - a_c - a_r \leq \beta$, and $e_r \geq a_r$ gives $M - a_c \leq \beta$. Afterwards each core arrival raises n_{late} and a_c together. The lead bound is the caller's promise. For (L4), each wrapper costs c_{wrap} plus services, and each direct exchange c_{dx} . *The rest of law (R6)*. The blank returns to z_P . The wrapper and the direct exchange change nothing outside $\text{cells}(P) \cup \{e_P\}$, net. The work grows by the length of the path, by (L4). $\square$

Lemma 5.9 (laws). *With states and validity as in Definitions 5.3 and 5.4, P satisfies laws (R1)–(R5).*

Proof. (R1): this is (I0). (R2): shown after Definition 5.4. (R3): each condition involving β ((L1), (L3)) only weakens as β grows, and the children's allowances do not depend on β . (R4): every condition refers to the contents of $\text{cells}(P)$ (queues, hub cells, children), and the children's validity depends only on their own cells, which lie in $\text{cells}(P)$. (R5): let π fix 0 and every non-helper of P , in particular all core tiles. Every condition refers to tiles only through “core source”, “core final” and “token”, and π maps tokens to tokens, so the conditions are unchanged. The non-helpers of Q_j are core tiles of P , so π fixes them, and (R5) for Q_j gives a new valid state of Q_j . All choices P makes (the spoke to serve, the tile for a direct exchange) depend only on roles and the ledger, so the strategy carries over to the relabelled state. $\square$

5.8. Finishing.

Lemma 5.10 (finish). *Let the state be valid with allowance β , with $a_c = M$, $a_r = r$ and the blank at z_P . Then P can finish: a blank-returning path makes every home of P hold its goal and changes no cell outside $\text{cells}(P)$. Its length, plus ω , is at most*

$$\begin{aligned} \sum_j k'_j \text{cost}_j + \sum_j (\omega(\varsigma_j) + \text{finish}_j) + \sum_j (2|A_j| + 2|X_j|) \\ + n_{\text{req}} c_{\text{wrap}} + n_{\text{dx}} c_{\text{dx}} + O(|\text{connector}|) + O(s_P r) + O(s_P), \end{aligned}$$

where $k'_j = k_j + |Y_j|$ and finish_j is the length of child j 's finish.

Proof. All sources of P have been handed out (Lemma 3.2), so every spoke is retired and, by (I2), no core source of P remains anywhere. Between requests the carrier is a token, by (I2) and (I3). *Drain.* Walk to w . For each spoke (including spokes with $\text{pop}_j = 0$), perform $|Y_j|$ services with the (token) carrier. These keep the routing conditions (Lemmas 5.7 and 5.6) and raise dr_j to $|Y_j|$; the $\dagger$ items are not needed during the finish. Each one moves Y_j one place towards Q_j , so afterwards every core final of P that was in Y_j has entered Q_j . Now every core final is in a child ((I3)), so by (I5) child j has received in_j goals. Each $\text{in}_j \leq \text{pop}_j$, and $\sum_j \text{in}_j = a_c = M = \sum_j \text{pop}_j$, so $\text{in}_j = \text{pop}_j$. Child j 's counters $a_c^{(j)}, a_r^{(j)}$ count distinct core and reserve goals, so $a_c^{(j)} \leq |\text{Core}(Q_j)|$ and $a_r^{(j)} \leq |\text{Res}(Q_j)|$; their sum is pop_j , so both are equalities, as the child's finish requires. *Children.* For each j , walk from w along A_j and X_j to z_j , let Q_j finish, and walk back (Lemma 2.2). Its homes now hold their goals, and nothing outside Q_j has changed. *Own homes.* Walk back to z_P . Every reserve goal of P is now in $\text{cells}(P)$ (since $a_r = r$) but not in a child home (those hold child goals), so it lies in $\text{Res}(P) \cup \text{Spare}(P)$. Apply Lemma 2.5 in $\mathcal{S}_P$ with $\mathcal{R} = \text{Res}(P) \cup \text{Spare}(P) \cup \{z_P\}$ and $Z = \text{Res}(P)$; the six markers give the

room needed. We call this the *final placement*. Children's homes and e_P are untouched. The bound on the length is (L4) plus these costs. $\square$

5.9. The budget.

Theorem 5.11. *Let P be a star layout whose children are regions with budgets $(\text{base}_j, \text{rate}_j)$. Then P , with the strategy above, is a region with*

$$\begin{aligned} \text{base}_P &= \sum_j \text{cost}_j (\text{pop}_j + |X_j| + |Y_j|) + J \text{cost}_{\max} + \sum_j (\text{base}_j + 2|A_j| + 2|X_j|) \\ &\quad + \text{rate}_{\max} \sum_j (3 + |X_j| + |Y_j|) + 2(M + r) c_{\text{wrap}} + 2r c_{\text{dx}} \\ &\quad + O(|\text{connector}|) + O(s_P r) + O(s_P), \\ \text{rate}_P &= (\text{cost}_{\max} + \text{rate}_{\max}) \mathcal{H}_J + c_{\text{dx}}, \end{aligned}$$

where $\text{cost}_{\max} = \max_j \text{cost}_j$ and $\text{rate}_{\max} = \max_j \text{rate}_j$.

Proof. Laws (R1)–(R5) hold by Lemma 5.9, and (R6) by Lemma 5.8. For (R7), bound the terms of Lemma 5.10 using validity. By Lemma 5.5(3) with $\text{dr}_j = 0$ and Lemma 5.5(2),

$$\sum_j k'_j \text{cost}_j \leq \sum_j \text{cost}_j (\text{pop}_j + |X_j| + |Y_j|) + \text{cost}_{\max} (J + \beta \mathcal{H}_J).$$

Child j is valid with allowance β_j , so by (R7) for it, $\omega(\varsigma_j) + \text{finish}_j \leq \text{base}_j + \text{rate}_j \beta_j$. By Lemma 5.5(2),

$$\sum_j (\text{base}_j + \text{rate}_j \beta_j) \leq \sum_j \text{base}_j + \text{rate}_{\max} (J + \beta \mathcal{H}_J + \sum_j (|X_j| + |Y_j| + 2)).$$

By (L3), $n_{\text{req}} \leq 2(M + r)$ and $n_{\text{dx}} \leq a_r + e_r + n_{\text{late}} \leq 2r + \beta$. Collecting the terms without β gives base_P , and the coefficient of β is $(\text{cost}_{\max} + \text{rate}_{\max}) \mathcal{H}_J + c_{\text{dx}}$. $\square$

6. THE CORNER TREE

Fix $b = 16$, so $J = b^2 = 256$ and $\mathcal{H}_J < 7$. Put $W = 2b + 2$ and $M_0 = 3b^2 + 4$. All constants below may depend on b . Fix a constant $T_{\min} \geq 2b + 4$, and let $T_{\text{big}} = M_0 + W + 12 + bT_{\min}$. This is all we need of $T_{\min}$: it gives leaves a body with at least 6 cells per row and at least one row, it gives $T_{\min} \geq b$ and $T' > 2b$ below, and every star has $T \geq T_{\text{big}} \geq W + M_0 + 12$. The constants $T_{\min}$, then Λ_r , then Λ_c are chosen in this order.

6.1. Nodes. A *placed square* of side T is a square $\mathcal{S}$ of the board together with a bijection from $[0, T]^2$ to $\mathcal{S}$ of the form $(i, i') \mapsto (k_0 \pm i, k'_0 \pm i')$, with a fixed choice of each sign. Its inverse gives *local coordinates* on $\mathcal{S}$. Since the map only translates and reflects, adjacency and distances in local coordinates are those of the board. A *corner node* (or simply *node*) of side $T \geq T_{\min}$ with entry column $\kappa \in [0, b)$ is built inside a placed square of side T . The *corner* o of the node is the cell with local coordinates $(0, 0)$. The first W rows form the *band*, and the remaining rows form the *body*. The node's cells are exactly its body, and its entry pair is $e = (2\kappa, 0)$, $z = (2\kappa + 1, 0)$ in the band.

Leaves. If $T < T_{\text{big}}$ the node is a leaf (Proposition 4.1). Its spare cells are six cells of the first body row, its core homes are the rest of the body, and it has no reserve homes.

Stars. If $T \geq T_{\text{big}}$, let $T' = \lfloor (T - M_0)/b \rfloor \geq T_{\min}$. The node is the following star layout (all coordinates local). See Figure 1 for a sketch.

- *Slots.* For $0 \leq \xi, \kappa' < b$ the slot (ξ, κ') is the square of side T' with corner $(\bar{r}_\xi, \bar{c}_{\kappa'}) = (W + 4 + \xi T', M_0 + \kappa' T')$. It holds the child $j = b\xi + \kappa'$, a node of side T' with entry column κ' , placed with the same signs, so that its local coordinates are those of P shifted

by $(\bar{r}_\xi, \bar{c}_{\kappa'})$. So $e_j = (\bar{r}_\xi + 2\kappa', \bar{c}_{\kappa'})$ and $z_j = (\bar{r}_\xi + 2\kappa' + 1, \bar{c}_{\kappa'})$. Since $M_0 \geq W + 4$, the slots fit in the body.

- *Hub*. The hub square has side $h = M_0 + 5$ and corner $(W, 0)$. It contains the hub cells $w = (W, M_0)$, $K, u, v, u' = (W+1, M_0+1), \dots, (W+1, M_0+4)$ and the H_j below. It may also contain some cells of the slots nearest the corner, which is harmless (Definition 5.1). The markers are $(W+1, M_0+5+i)$ for $0 \leq i < 6$. The *margin* is the set of cells in columns $0, \dots, M_0 - 1$ and rows $W+3, \dots, T-1$.
- *Spoke $j = b\xi + \kappa'$* . Let $\gamma_j = M_0 - 3(j+1) \geq 1$ and $y_j = \bar{r}_\xi + 2\kappa'$. Then $x_j = (W+2, \gamma_j)$ and $H_j = (W+2, \gamma_j+1)$. The export queue runs down column γ_j from row $W+3$ to row y_j+1 , then right along row y_j+1 up to z_j . The delivery queue runs from e_j left along row y_j to column γ_j+1 , then up column γ_j+1 to row $W+3$. The access walk runs from w left along row W to column γ_j , then down to x_j .
- *Connector*. From z down column 0 to row W , then right along row W to w .

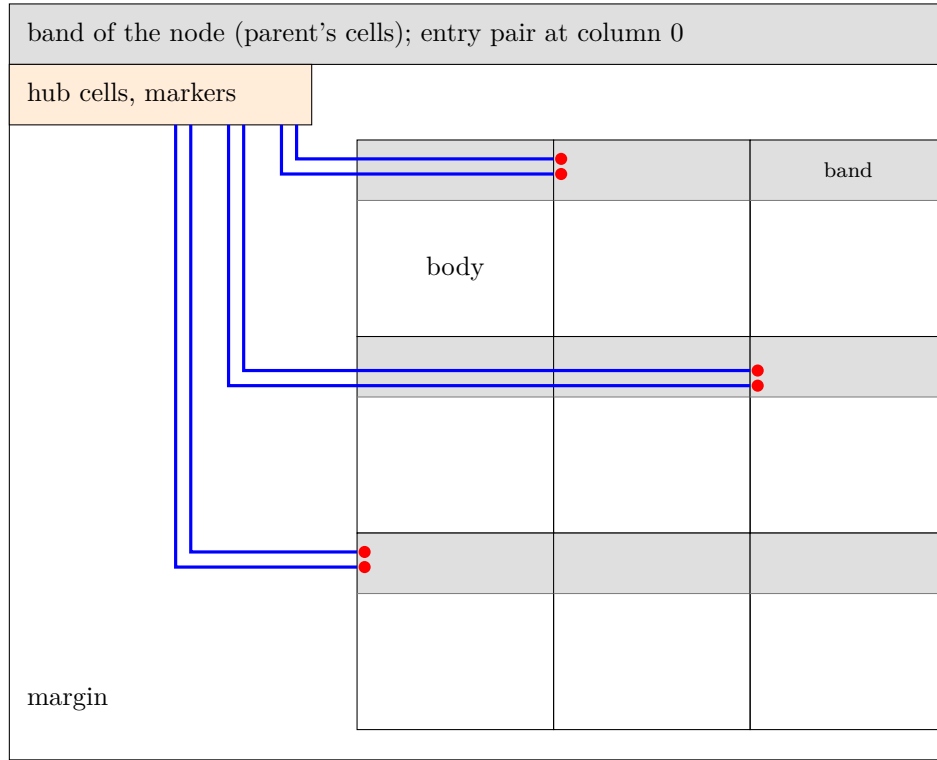


FIGURE 1. A corner node, not to scale, with 3×3 instead of $b \times b$ slots and three of the J spokes drawn. Each slot holds a child: its top W rows (shaded) are the child's band, the rest is the child's body. Each spoke's two lanes (blue; the outer one is the export lane, the inner one the delivery lane) leave the hub, run down two margin columns of their own, then turn right along two rows of their own inside the bands of their slot row, and end at the child's entry pair (red), in the first column of the child's square. Later spokes use columns further left and rows further down (a child further right has its entry pair two rows lower in the band), so no two lanes meet, and no lane enters a child's body. With 3×3 slots there would be 9 spokes and 18 lanes; the figure shows 6 of them.

Lemma 6.1 (corner layout). *This is a star layout (Definition 5.1) whose body is the node's body. Moreover:*

- (1) $|X_j| + |Y_j| \leq 2\Delta_j + 4b$, where $\Delta_j = \bar{r}_\xi + \bar{c}_{\kappa'}$ is the local distance from o to the corner of slot j ; also $|X_j|, |Y_j| \leq 2T$;
- (2) $h, |A_j|$ and the connector are $O(1)$, so $\text{cost}_j = |X_j| + |Y_j| + O(1)$ and $c_{\text{wrap}} = O(1)$;
- (3) $r = |\text{Res}(P)| = O(T)$.

Proof. Disjointness. Spoke j uses the margin columns $\gamma_j, \gamma_j + 1$, which differ for different spokes. Its horizontal part uses rows $y_j, y_j + 1$ inside the band of slot row ξ , and y_j is strictly increasing in j (consecutive slot rows are $T' > 2b$ apart). The vertical part of a spoke $j' < j$ lies in columns to the right of $\gamma_j + 1$ and ends at row $y_{j'} + 1 < y_j$, so it does not meet the horizontal part of spoke j . Symmetrically, the vertical part of a spoke $j' > j$ lies in columns to the left of γ_j , and its horizontal part, which crosses columns $\gamma_j, \gamma_j + 1$, lies in rows $y_{j'} \geq y_j + 2$, below the end (row $y_j + 1$) of spoke j 's vertical part. Horizontal parts run through the bands of slots, never through bodies, so loops avoid all children. Loops avoid row $W + 1$ (where K, u, v, u' and the markers are) and row W (where w , the access walks and the connector run). The remaining conditions of Definition 5.1:

- A_j runs along row W and then down column $\gamma_j < M_0$ through rows $W + 1, W + 2$. So it meets L_j only at x_j (the loop's vertical parts start at row $W + 3$), and it avoids K, u, v (row $W + 1$, columns $> M_0$) and all children (rows $< W + 4$).
- The hub square contains w, K, u, v, u' and every $H_j = (W + 2, \gamma_j + 1)$. The markers lie in row $W + 1$, which no loop meets.
- The connector uses column 0 above row W and row W itself. It avoids the loops (columns ≥ 1 , rows $\geq W + 2$), the children (rows $\geq W + 4$), K, u, v (row $W + 1$) and $e = (2\kappa, 0)$, which lies above z .
- x_j, H_j, K, u, v, u', w and the markers lie in no child. The body (rows $\geq W$) excludes e, z .
- The wrapper square is the square of side $h_w = W + M_0 + 6$ at the node's corner. It contains e, z, w, K, u, v , and it fits in the node's square since $T \geq W + M_0 + 12$; so do the hub square and the markers.

The definitions allow two harmless overlaps, which only matter for paths that restore everything: a child's connector runs through the band of its slot, crossing the lanes of later spokes in the same slot row, and the hub square overlaps margin loops and slot cells. Requests and double transpositions only promise net restoration, so neither matters. *Lengths.* $|X_j| = (y_j - W - 1) + (\bar{c}_{\kappa'} - \gamma_j - 1) + 1$ and $|Y_j| = 1 + (\bar{c}_{\kappa'} - \gamma_j - 1) + (y_j - W - 3)$. Since $y_j \leq \bar{r}_\xi + 2b$, this gives (1). (2) is immediate. *Reserve homes.* Own cells are the margin ($O(M_0 T)$ cells), the four hub rows above the slots, the bands of the slots ($b^2 \cdot WT' \leq bWT$ cells), and the leftover strips to the right of and below the slots, of width less than $b + M_0$. The children's spare cells add $6J$. In total $O(T)$. $\square$

6.2. The cost recursion. For a node V with corner o_V let $D_V = \sum_{x \in \text{cells}(V)} d(o_V, x)$, and let $\text{dp}(T)$ be the depth of the tree below a node of side T : 0 for a leaf, and $\text{dp}(T') + 1$ for a star.

Theorem 6.2. *There are constants Λ_r, Λ_c such that every node V of side $T \geq T_{\min}$, for a goal board whose blank is not in the node's body, is a region with*

$$\text{rate}_V \leq \Lambda_r T, \quad \text{base}_V \leq 2D_V + \Lambda_c T^2 (\text{dp}(T) + 1).$$

Moreover $\text{dp}(T) \leq \log_b T$.

Proof. Induction on T . For a leaf, $\text{rate} = 0$ and $\text{base} = O(T^3)$; since $T < T_{\text{big}}$, this is at most a constant times T^2 , which Λ_c absorbs.

Let $V = P$ be a star with children Q_j of side $T' \leq T/b$. By induction and Theorem 5.11 it is a region. We use Lemma 6.1 throughout.

Rate. $\text{cost}_{\max} \leq 4T + O(1)$, $\text{rate}_{\max} \leq \Lambda_r T' \leq \Lambda_r T/b$ and $c_{\text{dx}} = O(T)$, so

$$\text{rate}_P \leq (4T + O(1) + \Lambda_r T/16) \cdot 7 + O(T) \leq \frac{7}{16} \Lambda_r T + O(T) \leq \Lambda_r T$$

once Λ_r is large enough. This uses $b > \mathcal{H}_J$: each level contracts the rate by the factor $\mathcal{H}_J/b < 1$.

Base. Let $\ell_j = |\text{cells}(Q_j)| \geq \text{pop}_j$. The terms of base_P in Theorem 5.11 are bounded as follows.

- $\sum_j \text{cost}_j \text{pop}_j \leq \sum_j \ell_j (|X_j| + |Y_j|) + O(T^2) \leq 2 \sum_j \ell_j \Delta_j + O(T^2)$, since $\sum_j \ell_j \leq T^2$.
- $\sum_j \text{cost}_j (|X_j| + |Y_j|) \leq J(4T + O(1)) 4T = O(T^2)$, and $J \text{cost}_{\max} + \sum_j (2|A_j| + 2|X_j|) = O(T)$.
- $\text{rate}_{\max} \sum_j (3 + |X_j| + |Y_j|) \leq (\Lambda_r T/b) J(4T + 3) = O(\Lambda_r T^2)$.
- $2(M + r)c_{\text{wrap}} + 2r c_{\text{dx}} + O(T(r + 6)) = O(T^2)$, since $s_P = T$.
- $\sum_j \text{base}_j \leq \sum_j 2D_{Q_j} + J\Lambda_c T'^2 (\text{dp}(T') + 1) \leq 2 \sum_j D_{Q_j} + \Lambda_c T^2 \text{dp}(T)$, since $bT' \leq T$.

Hence $\text{base}_P \leq 2 \sum_j (D_{Q_j} + \ell_j \Delta_j) + \Lambda_c T^2 \text{dp}(T) + \Lambda_0 T^2$, where Λ_0 depends on b and Λ_r but not on Λ_c .

Telescoping. A cell x of child j has local coordinates $(\bar{r}_\xi + r', \bar{c}_{\kappa'} + c')$ with $r', c' \geq 0$ its coordinates in the child. So $d(o_P, x) = \Delta_j + d(o_{Q_j}, x)$. The children's cells are disjoint cells of P , so

$$D_P \geq \sum_j \sum_{x \in \text{cells}(Q_j)} (\Delta_j + d(o_{Q_j}, x)) = \sum_j (\ell_j \Delta_j + D_{Q_j}).$$

Therefore $\text{base}_P \leq 2D_P + \Lambda_c T^2 (\text{dp}(T) + 1)$, provided $\Lambda_c \geq \Lambda_0$ and Λ_c also covers the leaves.

Depth. $T' \leq T/b$, and leaves have $T \geq T_{\min} \geq b$, so $\text{dp}(T) \leq \log_b T$. $\square$

Informally: each tile whose home is in a node travels down the delivery lanes of all the ancestors of its home, and each source leaves through their export lanes. At each level the lanes cost the distance to the next corner, up to $O(1)$, and these distances add up along the chain of corners. The bound charges about $2D_V$ for this, which is why the factor 2 appears.

7. THE ROOT AND THE MAIN THEOREM

7.1. The central root. Let n be large and $T = \lfloor (n - 5)/2 \rfloor$, so the square $\mathcal{S}^* = [0, N^*)^2$ with $N^* = 2T + 5$ fits on the board, with $N^* \in \{n - 1, n\}$. Let $\chi = T + 2$, so the cell (χ, χ) is the centre of $\mathcal{S}^*$. For each $j \in \{1, 2, 3, 4\}$ with sign pair $(\iota_1, \iota_2) \in \{\pm 1\}^2$, the *reflection* ρ_j maps $(i, i') \mapsto (\chi + \iota_1(i - \chi), \chi + \iota_2(i' - \chi))$. It maps $\mathcal{S}^*$ onto itself, fixes the centre and preserves adjacency and distances. We describe spoke j in j -coordinates: the cell with j -coordinates (i, i') is the board cell $\rho_j(i, i')$. Put $\mu = \chi + 1$ and $\nu = \chi + 3$. In j -coordinates:

- child j is a corner node of side T with entry column 0, in the placed square $(i, i') \mapsto \rho_j(\nu + i, \nu + i')$, $0 \leq i, i' < T$, which is the square $[\nu, N^*)^2$ in j -coordinates. So $e_j = (\nu, \nu)$ and $z_j = (\nu + 1, \nu)$;
- $x_j = (\mu, \mu)$ and $H_j = (\mu, \mu + 1)$. The export queue is $(\mu + 1, \mu), (\mu + 2, \mu), (\mu + 3, \mu), (\nu + 1, \mu + 1), z_j$, and the delivery queue is $e_j, (\nu, \mu + 1), (\mu + 1, \mu + 1)$. The access walk is $(\chi, \chi), (\mu, \chi), x_j$.

In board coordinates, the hub square is the 7×7 square with corner $(\chi - 2, \chi - 2)$. The hub cells are $w = (\chi, \chi)$, $K, u, v, u' = (\chi, \chi + 1), \dots, (\chi, \chi + 4)$ and the four H_j . The entry pair is $e^* = (\chi, \chi - 2)$, $z^* = (\chi, \chi - 1)$, the connector is $z^* \rightarrow w$, and the markers are $(\chi, \chi + 5 + i)$, $0 \leq i < 6$. The body is the whole board except e^*, z^* . The wrapper square is the hub square, and the square of the root is the whole board.

Lemma 7.1. *This is a star layout with $J = 4$ in which every loop, access walk and connector has length $O(1)$, and $r = O(n)$.*

Proof. Call row χ and column χ of $\mathcal{S}^*$ the *middle row* and *middle column*. A cell whose two j -coordinates are at least $\mu = \chi + 1$ lies, on the board, strictly on one side of the middle row and of the middle column, in the quarter determined by (ι_1, ι_2) . The loop, x_j , H_j and the child of spoke j consist of such cells, so different spokes are disjoint, and none of them meets the middle row or the middle column. The middle row holds w, K, u, v, u' , the entry pair and the markers. The access walk passes through $\rho_j(\mu, \chi)$, which is on the middle column but on no loop. The hub square, which is also the wrapper square, contains w, K, u, v, u' , the entry pair and every $H_j = \rho_j(\mu, \mu + 1)$, which has board coordinates $(\chi \pm 1, \chi \pm 2)$. The access walk meets its loop only at x_j and avoids K, u, v . The connector consists of z^* and w alone. The markers lie on the middle row. The own cells are the central cross of width 5, the four children's bands ($O(T)$ cells each), and possibly the last row and column. So $r = O(n)$. $\square$

Let τ be the target board, with blank at the corner $\theta = (n - 1, n - 1)$, and let the goal G' be τ with the contents of z^* and θ exchanged. So G' has its blank at z^* , which is in no cell of P^* or of any node below it. Hence $G'(x) \neq 0$ for every home x of every node, as frames require. The root P^* is this star layout with goal G' .

Lemma 7.2. *P^* is a region with $\text{base}_{P^*} + \text{rate}_{P^*} \leq 8T^2(T - 1) + 4\Lambda_c T^2(\log_b T + 1) + O(n^2)$.*

Proof. Apply Theorem 5.11 with $J = 4$, $\mathcal{H}_4 < 3$. Everything except the children's bases is $O(n^2)$: $\text{cost}_j = O(1)$, $\text{pop}_j \leq T^2$, $|X_j|, |Y_j| = O(1)$, $\text{rate}_{\max} \leq \Lambda_r T$ (Theorem 6.2), $c_{\text{dx}} = O(n)$, $r = O(n)$, and $\text{rate}_{P^*} = O(n)$. By Theorem 6.2, $\text{base}_j \leq 2D_{Q_j} + \Lambda_c T^2(\log_b T + 1)$, and $D_{Q_j} \leq \sum_{0 \leq r', c' < T} (r' + c') = T^2(T - 1)$. $\square$

7.2. Driving the root.

Lemma 7.3 (driver). *Let R be a region with entry pair (e_R, z_R) , on a board with the blank at z_R , no blank in $\text{cells}(R)$, and every core and reserve home of R holding a goal tile of R . Then a path of length at most $\text{base}_R + \text{rate}_R$ makes every home of R hold its goal, returns the blank to z_R , and changes no cell outside $\text{cells}(R) \cup \{e_R\}$.*

Proof. Start R (allowance 0), and raise its allowance to 1 by (R3). Its sources are the tiles in its homes, all goal tiles. Make a helper request with the tile at e_R (which is not the blank). The output is a source, hence a goal tile of R , and it is now outside R . Feed it straight back as a core or reserve arrival according to whether its home is a core or a reserve home. Each such arrival again outputs a source, while any remain, and we feed it back too. The lead is 1 after the first request and stays 1 while sources remain, so allowance 1 suffices. After the helper request and $M + r$ goal arrivals, every source has been handed out and received back, so $a_c = M$ and $a_r = r$, and R finishes. The total work is at most $\text{base}_R + \text{rate}_R \cdot 1$. $\square$

7.3. Proof of Theorem 1.1.

Proof. Solving to the target. Let B be connected to τ .

- (1) Move the blank to z^* (Lemma 2.9): at most $2n$ moves.
- (2) Let $\mathcal{K}$ consist of 0 and the seven tiles $G'(x)$ for x a marker or e^* , and let $\mathcal{R} = \text{markers} \cup \{e^*, z^*\}$. Gather $\mathcal{K}$ into $\mathcal{R}$ (Lemma 2.8): $O(n)$ moves. Now every home of P^* (every cell other than the markers, e^* and z^*) holds a tile not in $\mathcal{K}$, that is, a goal tile of P^* .
- (3) Drive P^* (Lemma 7.3): at most $8T^2(T - 1) + 4\Lambda_c T^2(\log_b T + 1) + O(n^2)$ moves. Afterwards the board agrees with G' , hence with τ , outside $\mathcal{R} \cup \{\theta\}$, and the blank is at z^* .
- (4) Move the blank to θ (Lemma 2.9): at most $2n$ moves, changing at most $2n$ cells.

- (5) The board now differs from τ on at most $2n+9$ cells, has its blank at θ , and is connected to τ . Lemma 2.7 finishes in $O(n^2)$ moves.

Since $2T \leq n$, $8T^2(T-1) \leq n^3$. The total is at most $n^3 + O(n^2 \log n)$.

Arbitrary pairs. Let B and B' be connected. Move the blank of B' to θ by a path ζ of at most $2n$ moves, giving B'' . Let σ be the renaming of tiles that turns B'' into τ ; it fixes 0. Moves do not depend on tile names, so the renamed board $\sigma(B)$ is connected to $\sigma(B'') = \tau$. The path from $\sigma(B)$ to τ found above, applied to B , leads to B'' , and ζ reversed leads on to B' . The cost is at most $n^3 + O(n^2 \log n) + 2n$. $\square$

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